

Temperature Sensors

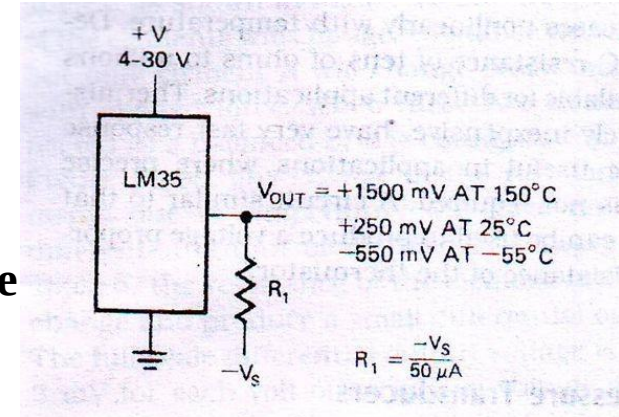
Semiconductor Temperature Sensors

Two types of semiconductor temperature sensors

- **Temperature-Sensitive Voltage Sources**
- **Temperature-Sensitive Current Sources**

Temperature-Sensitive Voltage Sources

- Example: LM35
- **Voltage output increases by 10 mV for each degree celsius in temperature increases**
- Output adjust to **0 V for 0° C.**
- Sensor gives output for temperature range of **-55° to +150°C**



Thermocouples

- A **thermocouple** is a **junction** between **two different metals** that **produces a voltage** related to a **temperature difference**.
- **When two different metals are put in contact ,a small voltage is produced between them—the Seebeck effect**
- Developed voltage depends on
 - **Type of metals used**
 - **Temperature**
- The developed **voltage increases between 7 and 75 μV** for **each degree celsius increase** in temperature- depending on the metal
- A thermocouple junction made of **iron and constantan** is called a type ‘J Thermocouple’
 - Temperature range : -184 to +760 degree Celsius
- Thermocouple junction can be made of **platinum and alloy of platinum-rhodium**
 - Temperature range : 0 to 1600 degree Celsius

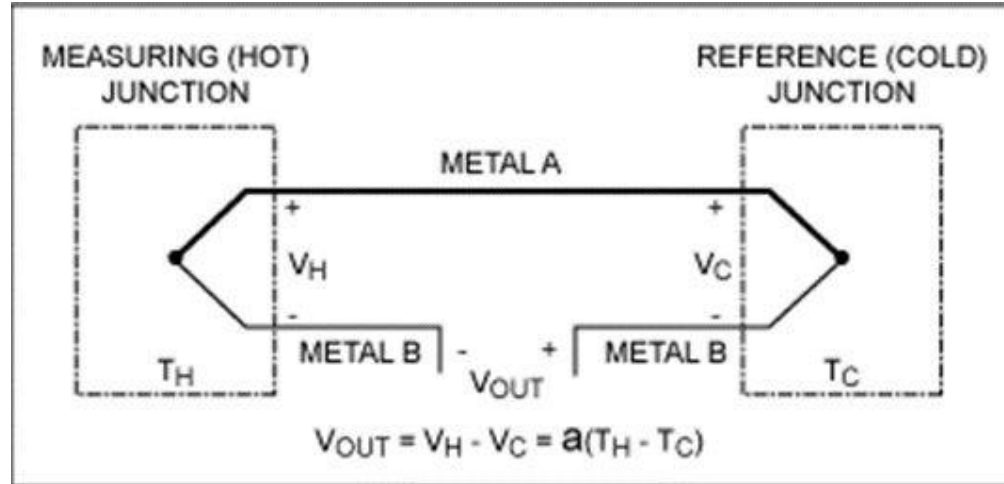
Thermocouples - 3 main problems

1. **Output is very small and need to be amplified** a great deal
2. **A reference junction of same metal must be connected in series** with the junction used to make the **measurement**
 - Because thermocouple reading is meaningless without comparing it to another, more accurate temperature sensor, **usually a reference**
 - The difference between the voltages across the two thermocouples goes as input to the amplifier
 - It's impossible to tell which thermocouple is causing a change in voltage if both **temperatures are changing**.
 - One end of the thermocouple—the hot junction—is thermally attached to the object to be measured
 - The other end of the thermocouple—the cold junction—is placed into an ice bath to hold it at a constant temperature

Thermocouples - 3 main problems

3. Output voltages do not change linearly with temperature

- Analog circuitry is needed which changes the gain of an amplifier according to the value of the signal



Thermocouple math problem

Suppose the measured output voltage between measuring junction and reference junction using thermocouple is 3.41mV. Reference junction temperature is 22°C. What will be the temperature of the measuring junction? Also find the Seebeck coefficient.

°C	0	1	2	3	4	5	6	7	8	9	10
Thermoelectric Voltage in mV											
0	0.000	0.039	0.079	0.119	0.158	0.198	0.238	0.277	0.317	0.357	0.397
10	0.397	0.437	0.477	0.517	0.557	0.597	0.637	0.677	0.718	0.758	0.798
20	0.798	0.838	0.879	0.919	0.960	1.000	1.041	1.081	1.122	1.163	1.203
30	1.203	1.244	1.285	1.326	1.366	1.407	1.448	1.489	1.530	1.571	1.612
40	1.612	1.653	1.694	1.735	1.776	1.817	1.858	1.899	1.941	1.982	2.023
50	2.023	2.064	2.106	2.147	2.188	2.230	2.271	2.312	2.354	2.395	2.436
60	2.436	2.478	2.519	2.561	2.602	2.644	2.685	2.727	2.768	2.810	2.851
70	2.851	2.893	2.934	2.976	3.017	3.059	3.100	3.142	3.184	3.225	3.267
80	3.267	3.308	3.350	3.391	3.433	3.474	3.516	3.557	3.599	3.640	3.682
90	3.682	3.723	3.765	3.806	3.848	3.889	3.931	3.972	4.013	4.055	4.096
100	4.096	4.138	4.179	4.220	4.262	4.303	4.344	4.385	4.427	4.468	4.509
110	4.509	4.550	4.591	4.633	4.674	4.715	4.756	4.797	4.838	4.879	4.920
120	4.920	4.961	5.002	5.043	5.084	5.124	5.165	5.206	5.247	5.288	5.328
130	5.328	5.369	5.410	5.450	5.491	5.532	5.572	5.613	5.653	5.694	5.735
140	5.735	5.775	5.815	5.856	5.896	5.937	5.977	6.017	6.058	6.098	6.138
150	6.138	6.179	6.219	6.259	6.299	6.339	6.380	6.420	6.460	6.500	6.540
160	6.540	6.580	6.620	6.660	6.701	6.741	6.781	6.821	6.861	6.901	6.941
170	6.941	6.981	7.021	7.060	7.100	7.140	7.180	7.220	7.260	7.300	7.340
180	7.340	7.380	7.420	7.460	7.500	7.540	7.579	7.619	7.659	7.699	7.739
190	7.739	7.779	7.819	7.859	7.899	7.939	7.979	8.019	8.059	8.099	8.138
200	8.138	8.178	8.218	8.258	8.298	8.338	8.378	8.418	8.458	8.499	8.539
210	8.539	8.579	8.619	8.659	8.699	8.739	8.779	8.819	8.860	8.900	8.940
220	8.940	8.980	9.020	9.061	9.101	9.141	9.181	9.222	9.262	9.302	9.343
230	9.343	9.383	9.423	9.464	9.504	9.545	9.585	9.626	9.666	9.707	9.747
240	9.747	9.788	9.828	9.869	9.909	9.950	9.991	10.031	10.072	10.113	10.153
250	10.153	10.194	10.235	10.276	10.316	10.357	10.398	10.439	10.480	10.520	10.561
260	10.561	10.602	10.643	10.684	10.725	10.766	10.807	10.848	10.889	10.930	10.971
270	10.971	11.012	11.053	11.094	11.135	11.176	11.217	11.259	11.300	11.341	11.382
280	11.382	11.423	11.465	11.506	11.547	11.588	11.630	11.671	11.712	11.753	11.795
290	11.795	11.836	11.877	11.919	11.960	12.001	12.043	12.084	12.126	12.167	12.209
300	12.209	12.250	12.291	12.333	12.374	12.415	12.457	12.499	12.540	12.582	12.624
310	12.624	12.665	12.707	12.748	12.790	12.831	12.873	12.915	12.956	12.998	13.040
320	13.040	13.081	13.123	13.165	13.206	13.248	13.290	13.331	13.373	13.415	13.457
330	13.457	13.498	13.540	13.582	13.624	13.665	13.707	13.749	13.791	13.833	13.874
340	13.874	13.916	13.958	14.000	14.042	14.084	14.126	14.167	14.209	14.251	14.293
350	14.293	14.335	14.377	14.419	14.461	14.503	14.545	14.587	14.629	14.671	14.713
360	14.713	14.755	14.797	14.839	14.881	14.923	14.965	15.007	15.049	15.091	15.133
370	15.133	15.175	15.217	15.259	15.301	15.343	15.385	15.427	15.469	15.511	15.554
380	15.554	15.596	15.638	15.680	15.722	15.764	15.806	15.849	15.891	15.933	15.975
390	15.975	16.017	16.059	16.102	16.144	16.186	16.228	16.270	16.313	16.355	16.397
400	16.397	16.439	16.482	16.524	16.566	16.608	16.651	16.693	16.735	16.778	16.820
410	16.820	16.862	16.904	16.947	16.989	17.031	17.074	17.116	17.158	17.201	17.243
420	17.243	17.285	17.328	17.370	17.413	17.455	17.497	17.540	17.582	17.624	17.667
430	17.667	17.709	17.752	17.794	17.837	17.879	17.921	17.964	18.006	18.049	18.091
440	18.091	18.134	18.176	18.218	18.261	18.303	18.346	18.388	18.431	18.473	18.516

Thermocouple math problem

Suppose the measured output voltage between measuring junction and reference junction using thermocouple is 3.41mV. Reference junction temperature is 22°C. What will be the temperature of the measuring junction? [Also find the seebeck](#)

We know, V_{out}
 $= V_h - V_c$
 $V_{out} = a (T_h - T_c)$
 $V_{out} = 3.41\text{mV}$
 $T_c = 22 \text{ degree C}$
 $T_h = ?$
 $a = ?$

From table,

When $T = 22 \text{ C}$, $V = 0.879 \text{ mV}$

So, $V_c = 0.879 \text{ mV}$

Thus, $V_{out} = V_h - V_c$

$$3.41 = V_h - 0.879$$

$$V_h = 4.289\text{mV}$$

So, From table, when $V = 4.289$ approx., $T = 104$ or 105 C So,
 $T_h = 105$ (4.289 close to 4.303 than 4.262)

$$\text{So, } a = 3.41 / (105 - 22) = 0.0411$$

Resistance Temperature Detector (RTD)

- **Resistance Temperature Detector** is another commonly used Temperature Sensor
- It is actually a **resistor** which **change value with a change in temperature**
- It consists of a **wire or a thin film of platinum or a nickel wire**
- Response of RTDs is **non-linear** but they have excellent
 - **Stability** (This is the sensor's ability to maintain a consistent output when a constant input is applied)
 - **Repeatability** (is the sensor's ability to give the same output or reading under repeated identical conditions.)
- It is useful for measures in the range of **-250 to +850° C**
- **The resistance of an RTD increases with an increase in temperature**
- Change in the **resistance** is **converted into a proportional voltage**
- **Used where precise temperature measurement is needed**

Thermistor

- Thermistors consists of **semiconductor materials**
- **Resistance decreases nonlinearly with temperature**
- Relatively inexpensive
- Very fast response time
- **Applicable where precise measurement is not required**

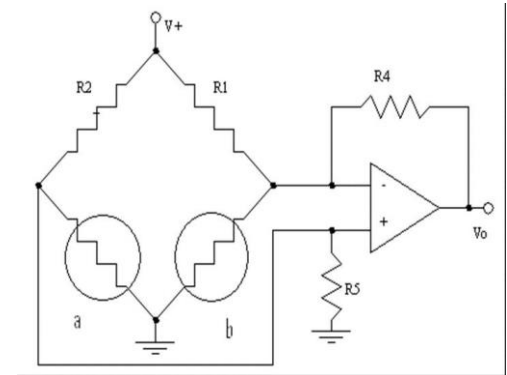
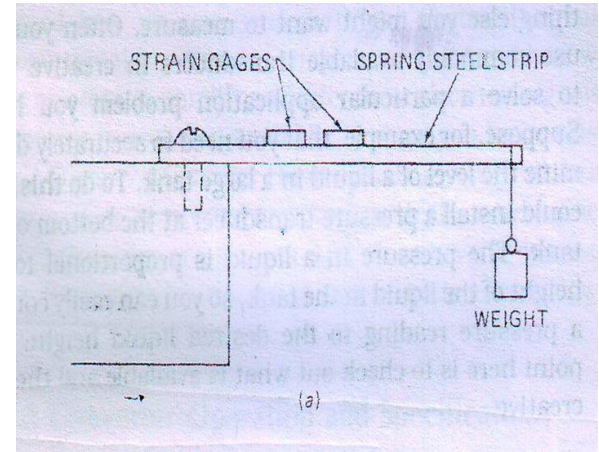
Force and Pressure Sensors

Strain Gages

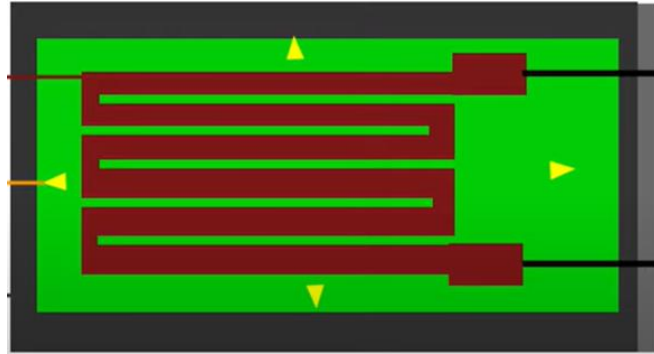
- It is a small **resistor** whose **value changes** when its **length is changed**
- It is made of **thin wire, thin foil or semiconductor material**
- **One end of a piece of spring steel is attached to a fixed surface**
- **A strain gage is glued on the top of the flexible bar**
- **The force or weight to be measured is applied to the unattached end of the bar**
- **As the applied force bend the bar, the strain gage is stretched, increasing its resistance**
- **Since the amount that the bar is bent is directly proportional to the applied force**
- **The change in the resistance will be proportional to the applied force**

Strain Gages

- If current is passed through the strain gage, **then the change in voltage across the strain gage will be proportional to the applied force**
- Unfortunately the **resistance of strain gage element also changes with the temperature**
- **Two Strain gage elements are mounted at right angles**
- **If the elements are connected in a balanced – bridge configuration, any change in the resistance due to temperature will have no effect on the differential output of the bridge**



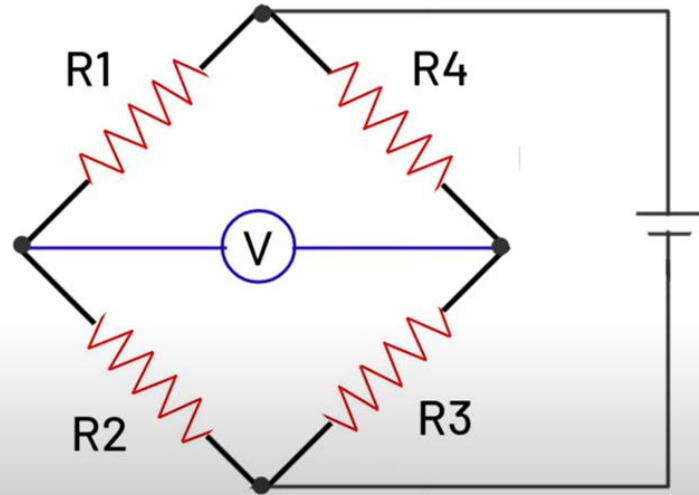
Strain Gages



Strain Gages

$$V = \frac{R3}{R3 + R4} - \frac{R2}{R1 + R2} \times V_{ex}$$

$$R2 / R4 = R1 / R3$$



Strain Gages

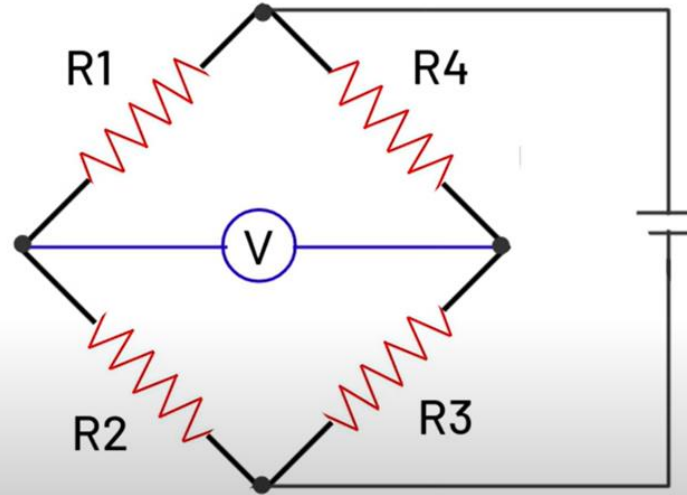
- gauge factor

$$K = \frac{\triangle R / R}{\triangle L / L}$$

K - gauge factor

$\triangle R$ - change in initial resistance

R - Initial resistance without strain



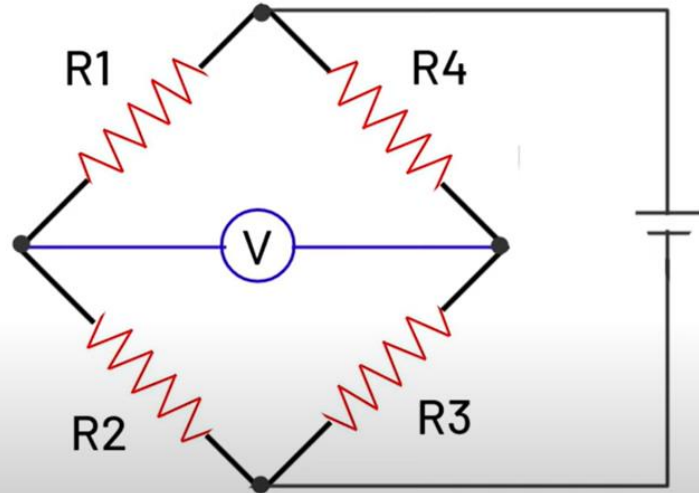
Strain Gages

As we know ,

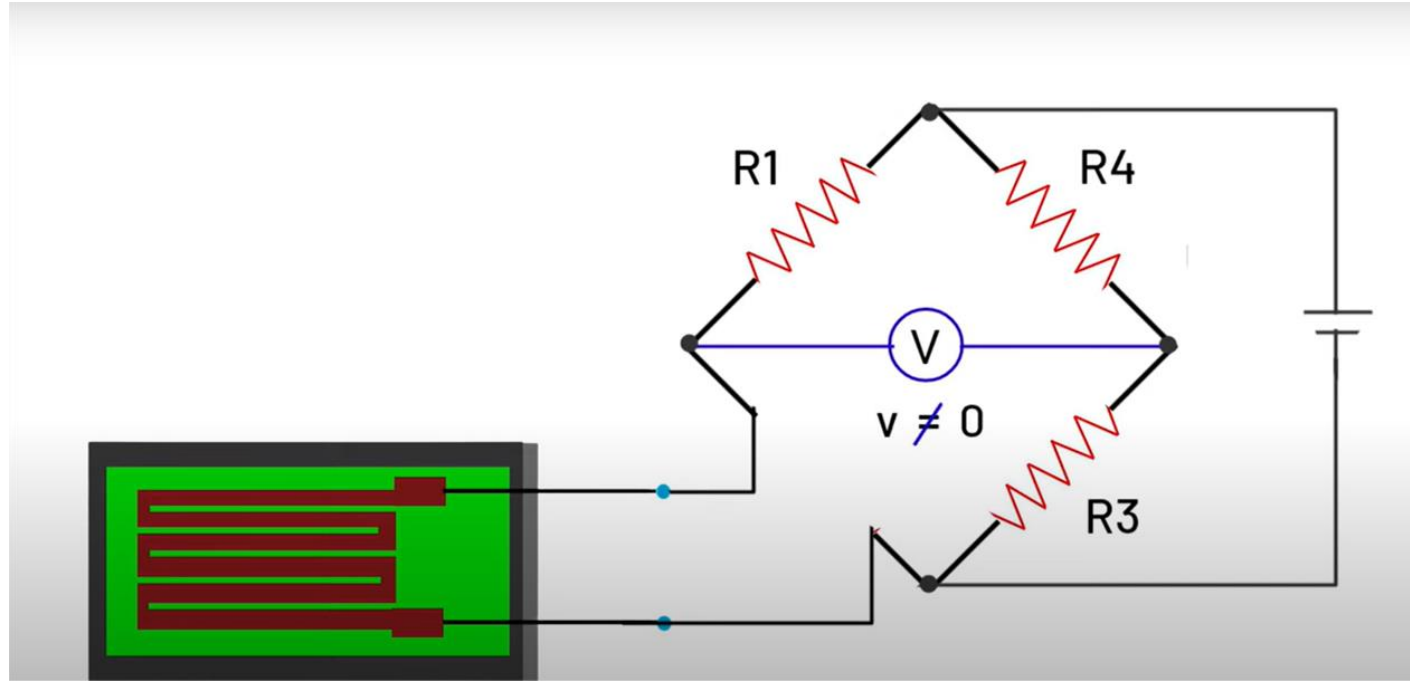
$$R = \frac{\rho L}{A}$$

R = is directly proportional to length (L)

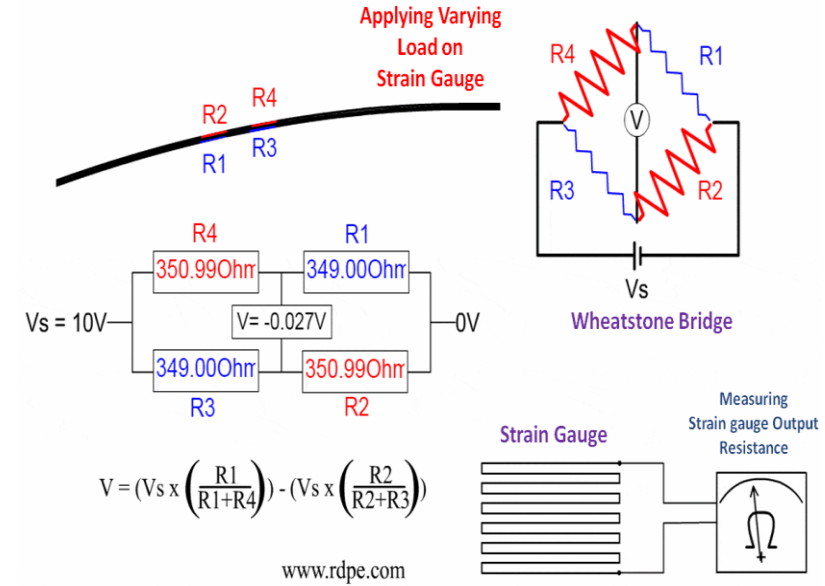
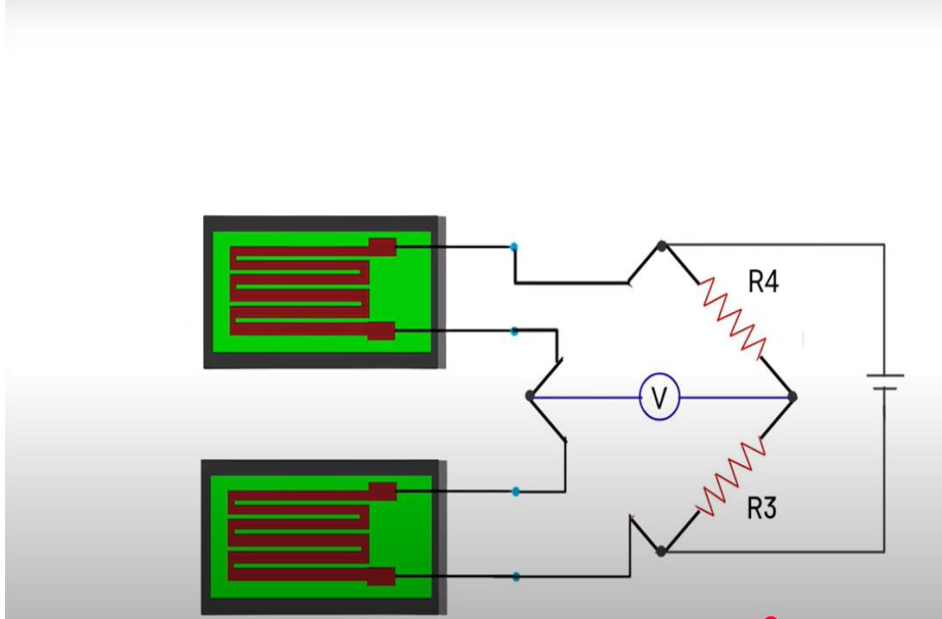
R = is inversely proportional to cross section area (A)



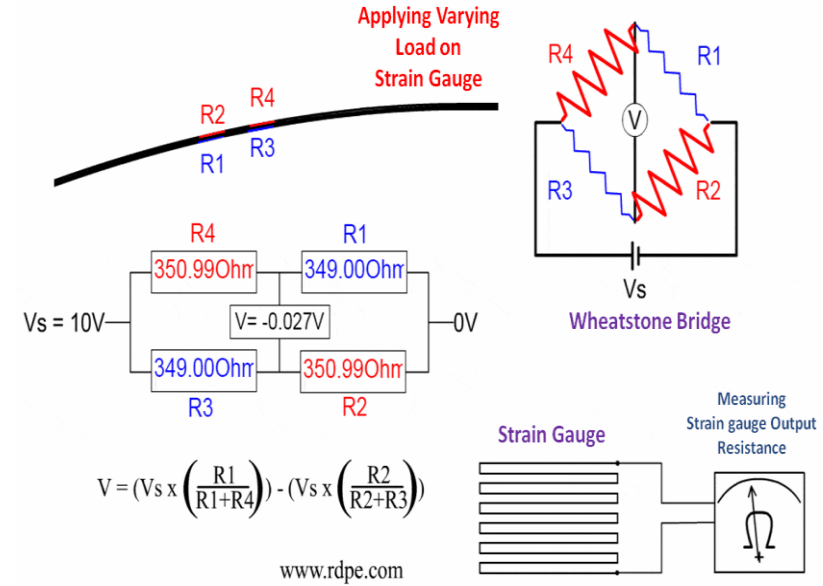
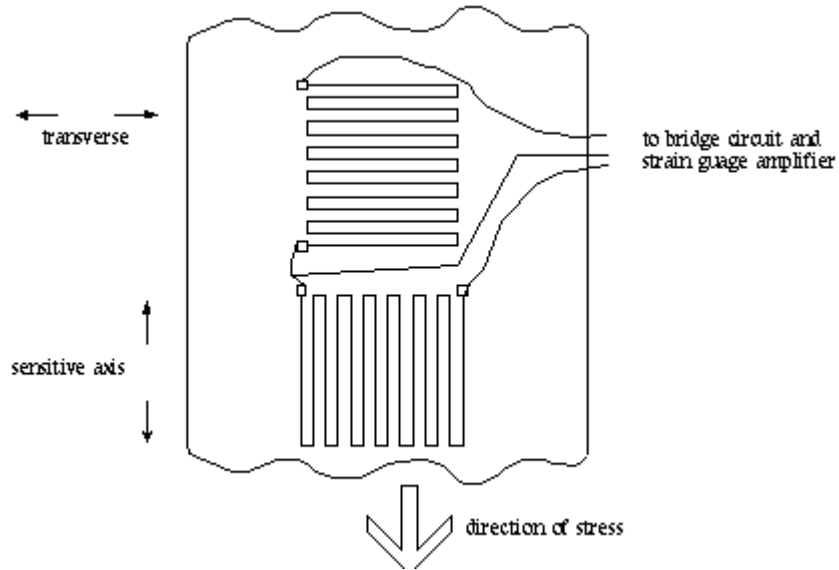
Strain Gages



Strain Gages



Strain Gages



Math

For a balance bridge: stretch resistor are 4Ω and compressed resistor are 2Ω .
 V_{in} is 5V. Find V_{out} .



**Military Institute of Science and
Technology Department of Computer Science and
Engineering**

Analog Interfacing and Industrial Control

Course Name: Computer Interfacing

Course Code: CSE-405

Book Reference

- ❏ Microprocessor and Interfacing-
Douglas V. Hall (Chapter 10)

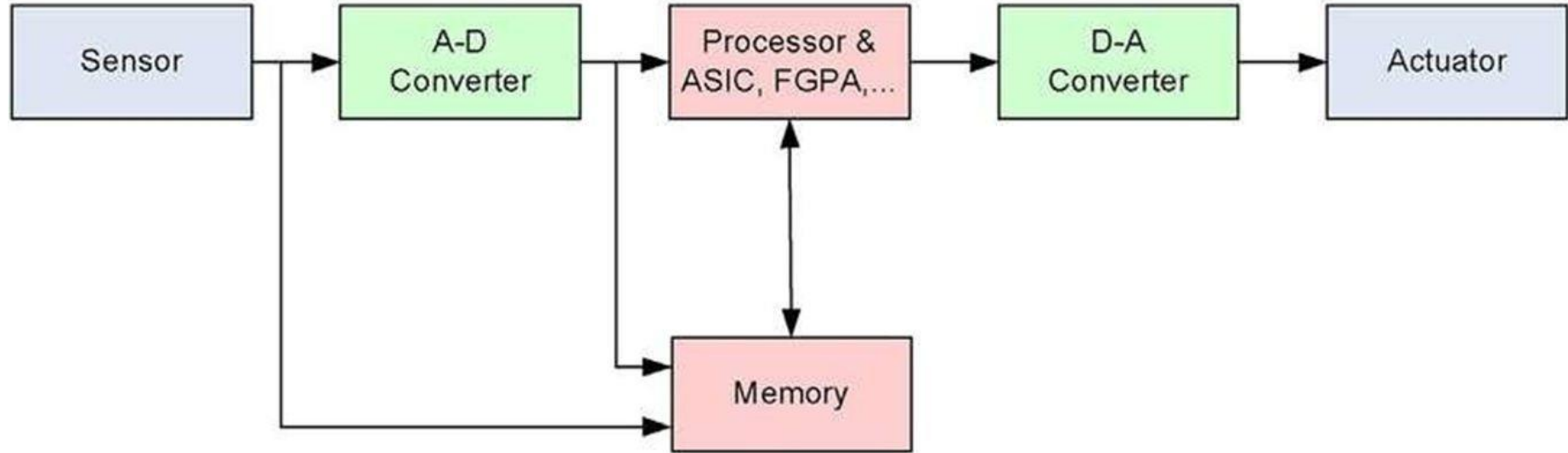
Embedded System

Introduction

- A computer hardware system having software embedded in it.
- A **microcontroller or microprocessor based system** which is designed to **perform a specific task**.
- either as an independent system or as a part of a large system.
- At the core is an integrated circuit designed to carry out computation for **real-time operations**.
- An embedded system has three components
 - It has hardware.
 - It has application software.
 - It has **Real Time Operating system (RTOS)** that defines the way the system works.
 - It sets the rules during the execution of application program.

<https://www.digi.com/blog/post/examples-of-embedded-systems>

Basic Structure of An Embedded System



Basic Structure of An Embedded System

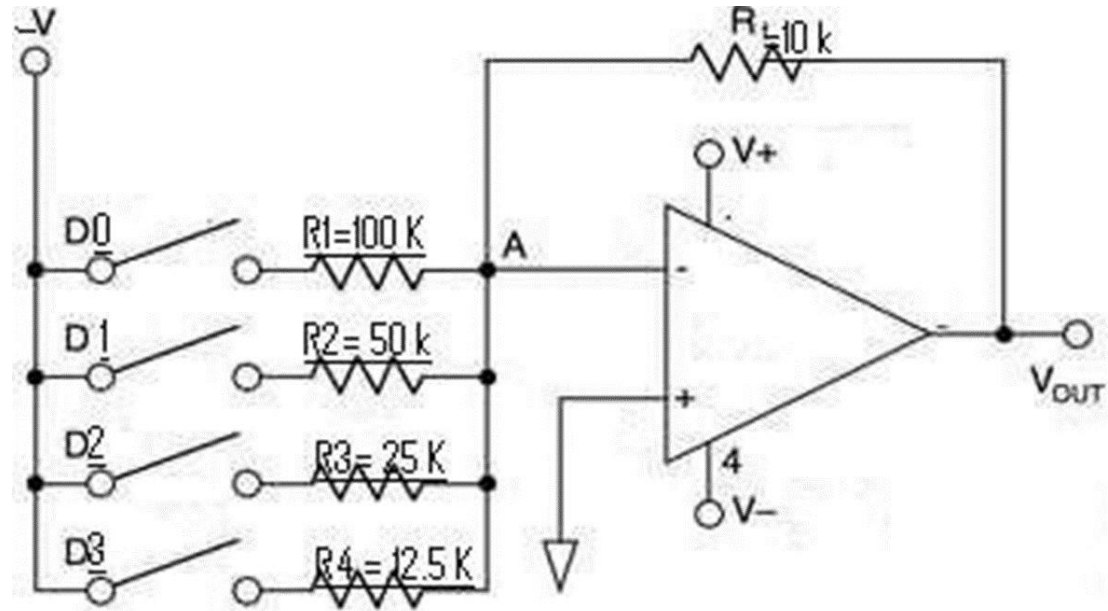
- **Sensor:** It measures the **physical quantity** and converts it to an electrical signal which can be read by an observer or by any **electronic instrument like an A/D converter**. A sensor stores the measured **quantity to the memory**.
- **A-D Converter:** An analog-to-digital converter **converts the analog signal sent by the sensor** into a digital signal.
- **Processor & ASICs:** **Processors process the data to measure the output** and store it to the memory.
- **D-A Converter:** A **digital-to-analog converter** converts the **digital data fed by the processor to analog data**
- **Actuator:** An actuator converts an electrical signal into physical action (movement, heat, pressure, rotation, etc.). It executes commands from the processor to control the physical world.

D/A Converter

D/A Converter: Operation

- The purpose of a Digital to Analog converter is to **convert a binary word to a proportional current or voltage.**
- In a simple 4 bit D/A converter , **inverting input of the circuit is referred to as the summing point.**
- When one of the switches is closed, **current will flow from -V to the summing point.**
- The equation of O/P voltage, $V_{\text{out}} = - R_f / R (V_{\text{ref}})$
- If we close the switch D0 , $V_{\text{out}} = - 10 / 100(-5) = 0.5 \text{ V}$
- If we close the switch D0 & D1 , $V_{\text{out}} = - 10 / 33.33(-5) = 1.5 \text{ V}$
- The point here is that, **binary weighted resistors produce currents which are summed by the op amp to produce a proportional output voltage**
- The heart of a **D/A converter is a set of binary weighted current sources which can be switched on or off according to a binary word applied to its input**

D/A Converter



D/A Converter: Characteristics

Resolution

- This is determined by the **number of bits in the input binary word**.
- A converter with **8 binary inputs** has **2^8 output levels**.
- Thus its resolution is **1 part in 256**.

Example:

- 12 bit converter has a resolution of 1 part in 2^{12} or 4096.
- Resolution is sometimes expressed in percentage.
- The resolution of an 8 bit converter can be expressed as a percentage is $(1/256)*100$ percent or 0.39 percent.

D/A Converter: Characteristics

Full scale output voltage

- The maximum output voltage of a converter will always have a value 1 least significant bit less than the named value.

Example:

- 12 bit, 10-V converter
- The value of 1 LSB = $(10\text{V})/4096 = 2.44\text{mV}$
- The highest voltage output for the converter when it is properly adjusted will then be : $(10 - 0.0024)\text{V}$ or 9.9976V .

D/A Converter: Characteristics

Example:

Q) An n bit D/A converter has **full scale output of 5V**. But actually it is **4.6875V**. what is the value of n ?

A) Actual output = full scale output - (full scale output / 2^n) $4.6875 = 5 - 5/(2^n) - 1$

- So, $n=4$
- value of 1 LSB = $(10V)/4096 = 2.44mV$
- The highest voltage output for the converter when it is properly adjusted will then be : $(10 - 0.0024)V$ or $9.9976V$.

D/A Converter: Characteristics

Accuracy

- The accuracy specification for a D/A converter is a comparison between the actual output and the expected output
- It is **specified as a percentage of the full scale output voltage or current**

Example

- If the converter has a output of **10V and ± 0.2 percent accuracy**
- Maximum error for any output will be **$0.002 * 10V$ or 20mV**

Linearity

- Linearity is the measure of **how much the output ramp deviates from a straight line** as the converter is stepped from no switches on to all switches on.
- **Ideally the deviation of the output from a straight line should be no greater than $\pm \frac{1}{2}$ the value of LSB to maintain overall accuracy**

D/A Converter: Characteristics

Settling Time

- When we change the binary word applied to the input of a converter , the output will change to the appropriate new value
- The output may overshoot the correct value, “ring” for a while before finally settling down to the correct value
- The time the output takes to get within $\pm \frac{1}{2}$ **LSB of the final value is called settling time**
- This specification is important, because if a converter is operated at too high frequency, it may not have time to settle to one value before it is switched to the next

D/A Converter: Interfacing to Microcomputers

- The inputs of the D/A circuit can be connected directly to the microcomputer output port.
- Thus interfacing an 8 bit converter involves simply connecting the inputs of the converter to an output port.
- For some application we need to interface a 12 bit converter to a microcomputer.
- If we are working with a system which has an 8 bit data bus
 - The first thought might be – to connect the lower 8 inputs of the 12 bit converter to one output port and upper 4 bits to another port
 - In that case, we could send the lower 8 bits with one write operation and the upper 4 bits with another write operation
 - Time between the two writes introduces a potential problem in this approach

D/A Converter: Interfacing to Microcomputers

Example:

- Suppose we want to change the pattern of a 12 bit converter from 0000 1111 1111 to 0001 0000 0000
- When we will write the lower 8 bits, the output will go from 0000 1111 1111 to 0000 0000 0000
- When we will write the upper 4 bits the output will go back up to the desired 0001 0000 0000
- The point here is that for the time between the two writes the output will go to an unwanted value
- In many systems this could be disastrous
- The cure of this problem is to put latches on the input lines
- The latches can be loaded separately and then strobed together to pass all 12 bits to the D/A converter at the same time

Thank You